Osterwalder–Schrader Axioms for the Interacting Universal Somatic Field: Reflection Positivity under Hopfield Coupling

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Abstract

The companion paper *The Universal Somatic Field as a Euclidean Quantum Field Theory* established that the free-field Universal Somatic Field (USF) satisfies all five Osterwalder–Schrader (OS) axioms, placing it within the rigorous framework of axiomatic Euclidean quantum field theory. The present paper develops the programme for proving OS axioms for the *interacting* USF — the theory with Hopfield coupling $\kappa > 0$ between field modes. The interaction term is quartic in field space and places the interacting USF in the same universality class as ϕ^4 field theory. We identify the key proof obligations: (i) a uniform lower bound on the Euclidean action; (ii) stability of reflection positivity (OS3) under κ-perturbation via the Glimm–Jaffe framework; (iii) an ultraviolet regularisation scheme compatible with the Hopfield weight matrix; and (iv) the thermodynamic limit. We state precise mathematical conjectures and outline a Lean 4 formalisation strategy that would close all obligations. Provisional constructions for obligations (i) and (ii) are given in the body; (iii) and (iv) remain open.

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1.1 1 Motivation and Context

The free-field USF was proved in the companion free-field paper to satisfy OS0–OS4 via its identification with the massive Gaussian Free Field. That result, though fundamental, covers only the linearised theory. The physical USF includes a Hopfield coupling $\kappa > 0$ that introduces non-linearity, attractor dynamics, and the pattern-completion behaviour that constitutes emotional memory.

The interacting theory is:

$$S_\kappa[\phi] = S_0[\phi] + \kappa\, V[\phi]$$

where S_0 is the free Euclidean action and $V[\phi]$ is the Hopfield interaction. In momentum space:

$$S_0[\phi] = \tfrac{1}{2} \int (p^2 + k^2)|\tilde{\phi}(p)|^2 \, \frac{d^4 p}{(2\pi)^4}$$

$$V[\phi] = -\tfrac{1}{2} \sum_{a,b} W_{ab} \int \phi(x)^a \phi(x)^b \, d^4 x$$

where W_{ab} is the Hopfield weight matrix and a, b index field components. For a single-component field and a scalar Hopfield weight W, this reduces to:

$$V[\phi] = -\tfrac{W}{2} \int \phi(x)^2 \, d^4 x = -\tfrac{W}{2} \int \frac{|\tilde{\phi}(p)|^2}{(2\pi)^4} \, d^4 p,$$

which is a mass renormalisation: $k^2 \to k^2 - \kappa W$. For the multi-component case, $V[\phi]$ contains quartic terms in the field components via the Hopfield energy function, placing it in the ϕ^4 universality class.

1.2 2 Proof Obligations

1.2.1 2.1 Obligation 1: Action lower bound (stability)

Conjecture 1. For κ sufficiently small (below the critical coupling $\kappa_c = k^2 / W_{\max}$ where $W_{\max}$ is the largest eigenvalue of W), the interacting action satisfies:

$$S_\kappa[\phi] \geq c_\kappa \, \|\phi\|_{H^1}^2 - C_\kappa$$

for constants $c_\kappa > 0$ and $C_\kappa < \infty$ depending on κ.

Proof strategy. Below κ_c, the effective mass $k^2 - \kappa W_{\max} > 0$ and the theory is in the Gaussian basin. The lower bound follows from completing the square in the action. At $\kappa = \kappa_c$ the theory undergoes a phase transition (spontaneous symmetry breaking), corresponding in USF terms to commitment to a trauma attractor.

1.2.2 2.2 Obligation 2: Stability of OS3 under perturbation

Conjecture 2. If μ_0 satisfies OS3 and V is a polynomial interaction bounded below, then the perturbed measure $d\mu_\kappa \propto e^{-\kappa V[\phi]}\, d\mu_0$ satisfies OS3 for κ in a neighbourhood of 0.

Proof strategy. This is the core result of the Glimm–Jaffe programme (Glimm & Jaffe, 1987, Chapter 8). For ϕ^4_4 theory it requires ultraviolet renormalisation (see §2.3). For the Hopfield-USF, the interaction V is quadratic in single-component fields but quartic in multi-component fields. The quadratic case admits a direct spectral argument; the quartic case requires the full Glimm–Jaffe machinery.

Current status. The quadratic (single-component, $V = -\frac{W}{2}\phi^2$) case is handled by mass renormalisation: $k_{\text{eff}}^2 = k^2 - \kappa W > 0$. OS3 holds by the free-field result with k_{eff}.

For the multi-component Hopfield theory the proof is *open*.

1.2.3 2.3 Obligation 3: Ultraviolet regularisation

The interacting theory requires a UV cutoff Λ and renormalisation. The USF Hopfield coupling introduces a natural scale through the weight matrix W, which in the neural network context is bounded (weights are learned from data). This suggests a natural UV regularisation:

$$W_{ab}(p) = W_{ab}^{(0)}\, f_\Lambda(p), \qquad f_\Lambda(p) = e^{-p^2/\Lambda^2}.$$

Obligation 3. Show that the $\Lambda \to \infty$ limit exists and yields a well-defined interacting measure satisfying OS3.

Current status. *Open.* The Gaussian damping makes each finite-Λ measure well-defined by the free-field argument. The limit requires uniform bounds in Λ, which in ϕ_4^4 theory require the full renormalisation group.

1.2.4 2.4 Obligation 4: Thermodynamic limit

Obligation 4. Show that the measures on bounded domains $\Lambda_L \uparrow \mathbb{R}^4$ converge to a well-defined infinite-volume measure.

Current status. *Open.* Standard for ϕ_2^4 (proved); open for ϕ_4^4, which is one of the Millennium Prize Problems (Yang–Mills mass gap is the gauge-theory analogue).

1.3 3 Lean 4 Formalisation Strategy

The formalisation for Obligations 1 and 2 (quadratic Hopfield case) would extend `USF_OSAxioms.lean` with:

```
-- Effective mass after Hopfield coupling (single-component, below κ_c)
noncomputable def k_eff (k κ W : ℝ) (hk : 0 < k) (hκ : 0 < κ) (hW : 0 < W)
    (hbelow : κ * W < k^2) : {m : ℝ // 0 < m} :=
  ⟨Real.sqrt (k^2 - κ * W), by positivity⟩


-- OS axioms for the single-component interacting USF below critical coupling
```

```
theorem interacting_USF_satisfies_OS_axioms_below_critical
    (k κ W : ℝ) [Fact (0 < k)] [Fact (0 < κ)] [Fact (0 < W)]
    (hbelow : κ * W < k^2) :
    SatisfiesAllOS (μ_GFF (k_eff k κ W).val) :=
  gaussianFreeField_satisfies_all_OS_axioms (k_eff k κ W).val
```

This closes Obligations 1–2 for the single-component case. The multi-component case requires new machinery not yet available in Lean 4's Mathlib.

1.4 4 Physical Interpretation of the Phase Transition

At $\kappa = \kappa_c = k^2/W_{\text{max}}$, the effective mass vanishes and the theory is critical. In USF terms this is the *trauma attractor transition*: below κ_c the field is in the Gaussian (healthy) phase with unique vacuum; at κ_c the correlation length diverges; above κ_c the symmetry breaks and the field settles into an attractor (trauma basin).

This phase structure matches the clinical phenomenology:

Phase	κ	Field state	Clinical analogue
Healthy	$\kappa < \kappa_c$	Gaussian, unique vacuum	Flexible emotional regulation
Critical	$\kappa = \kappa_c$	Diverging correlations	Threshold/tipping point
Trauma	$\kappa > \kappa_c$	Broken symmetry, attractor	Trauma fixation, hypervigilance

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1.5 5 Open Problems and Conclusion

The interacting USF presents a tractable research programme at the interface of constructive QFT and formal verification:

Established in the companion free-field paper: Free-field USF satisfies OS0–OS4 in its stated Lean 4 formalisation.

Closed (this paper): Single-component Hopfield USF below critical coupling satisfies OS0–OS4 via mass renormalisation.

Open: Multi-component Hopfield USF OS axioms require the full Glimm–Jaffe programme. This is mathematically hard (analogous to ϕ^4_4 theory) but physically motivated: it would give a rigorous foundation for the FM-HN model of emotional dynamics with multiple coupled modes.

Very long-term: The gauge-theory analogue — proving OS axioms for a non-Abelian gauge theory over the USF target space — connects to the Yang–Mills mass gap problem.

The present paper establishes the *proof programme* and closes the easiest case (single-component, subcritical). Full closure of the multi-component case is designated the central open problem of the SFT programme.

1.6 References

Glimm, J., & Jaffe, A. (1987). *Quantum physics: A functional integral point of view* (2nd edn). Springer.